Course: DSA 3050A SS 2026 End Semester Exam
Project Title: Strategic Healthcare Analytics: Predictive Modeling of Patient No-Shows
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Admission Number: 670035
GitHub Repository: [https://github.com/Calvin-Gacheru/Business-Intelligence-And-Visualization/tree/main/End%20Semester%20Project]

Introduction

Patient no-shows represent a significant operational challenge in healthcare administration, leading to idle clinical resources and delayed care. This report outlines the development of a complete Power BI analytical solution designed to address this inefficiency.

Using a raw dataset of over 110,000 medical records, this project executes rigorous data cleaning and transformation to build a robust relational model. The final interactive dashboard utilizes advanced DAX calculations to uncover demographic, geographic, and scheduling patterns correlated with missed appointments. The objective is to deliver data-driven insights and actionable recommendations that optimize operational efficiency and improve patient triage.

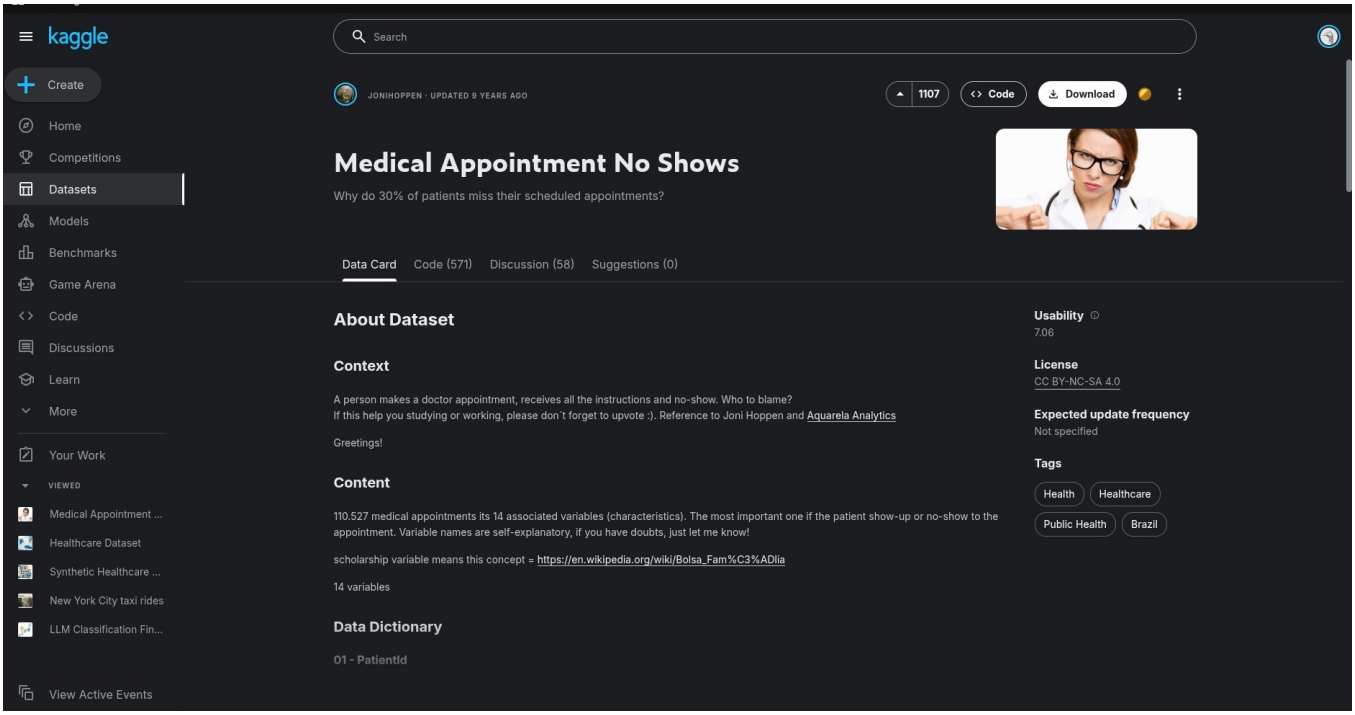
Part A: Data Acquisition and Understanding

1. Domain and Business Problem

- Domain: Healthcare Analytics.
- Business Problem: Patient no-shows cause idle clinical resources and operational bottlenecks. The objective is to analyze historical appointment data to identify demographic, geographic, and scheduling patterns that correlate with missed appointments, enabling targeted interventions and optimized scheduling.

2. Dataset Source

Source: Kaggle. The dataset is titled "Medical Appointment No-Shows (originally sourced from public medical records in Brazil).



3. Key Tables and Fields

The raw dataset is a single flat file containing 110,527 rows.

Key variables include:

- Identifiers: `PatientId`, `AppointmentID`
- Demographics: `Gender`, `Age`, `Neighbourhood`
- Timestamps: `ScheduledDay`, `AppointmentDay`
- Health Conditions: `Hipertension`, `Diabetes`, `Alcoholism`, `Handcap`
- Operational: `Scholarship` (welfare enrollment), `SMS_received`, `No-show` (target variable)

To fulfill the relational model requirement, this data will be transformed in Power Query into a star schema containing:

- DimPatient: `PatientId`, demographics, and health conditions.
- DimLocation: `Neighbourhood`.
- DimDate: Derived from the timestamp fields.
- FactAppointments: `AppointmentID`, `PatientId`, location/date keys, `SMS_received`, and `No-show` status.

4. Suitability for Advanced Power BI Analysis

This dataset is suitable for advanced Power BI analysis due to:

- Volume: Contains over 110,000 rows, exceeding the 9,000 minimum.
- Complexity: Includes categorical, numerical, boolean, and datetime variables.Transformation Needs: The raw data is messy. It requires text standardization, splitting datetime strings, handling invalid records (e.g., negative ages), and unpivoting/splitting columns to build the required relational tables.

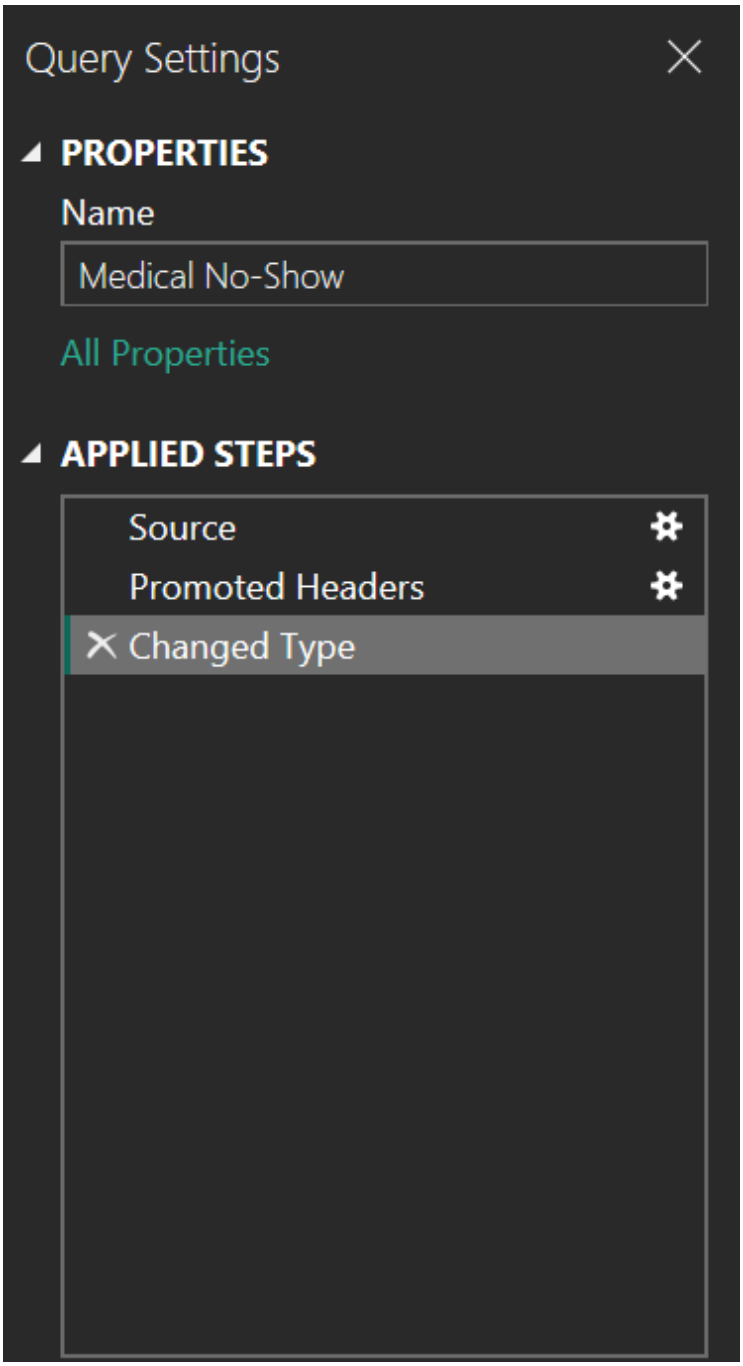
View of the raw data in Power Query Editor:

A	B	C	D	E	F	G	H	I	J	K	L	M	N
PatientId	AppointmentID	Gender	ScheduledDay	AppointmentDay	Age	Neighbourhood	Scholarship	Hipertension	Diabetes	Alcoholism	Handcap	SMS_received	No-show
29872499824296	5642903	F	2016-04-29T18:38:08Z	2016-04-29T00:00:00Z	62	JARDIM DA PENHA	0	1	0	0	0	0	No
558997776694438	5642503	M	2016-04-29T16:08:27Z	2016-04-29T00:00:00Z	56	JARDIM DA PENHA	0	0	0	0	0	0	No
4262962299951	5642549	F	2016-04-29T16:19:04Z	2016-04-29T00:00:00Z	62	MATA DA PRAIA	0	0	0	0	0	0	No
867951213174	5642828	F	2016-04-29T17:29:31Z	2016-04-29T00:00:00Z	8	PONTAL DE CAMBURI	0	0	0	0	0	0	No
8841186448183	5642494	F	2016-04-29T16:07:23Z	2016-04-29T00:00:00Z	56	JARDIM DA PENHA	0	1	1	0	0	0	No
95985133231274	5626772	F	2016-04-27T08:36:51Z	2016-04-29T00:00:00Z	76	REPUBLICA	0	1	0	0	0	0	No
733688164476661	5630279	F	2016-04-27T15:05:12Z	2016-04-29T00:00:00Z	23	GOIABEIRAS	0	0	0	0	0	0	Yes
3449833394123	5630575	F	2016-04-27T15:39:58Z	2016-04-29T00:00:00Z	39	GOIABEIRAS	0	0	0	0	0	0	Yes
56394729949972	5638447	F	2016-04-29T08:02:16Z	2016-04-29T00:00:00Z	21	ANDORINHAS	0	0	0	0	0	0	No
78124564369297	5629123	F	2016-04-27T12:48:25Z	2016-04-29T00:00:00Z	19	CONQUISTA	0	0	0	0	0	0	No
734536231958495	5630213	F	2016-04-27T14:58:11Z	2016-04-29T00:00:00Z	30	NOVA PALESTINA	0	0	0	0	0	0	No
7542951368435	5620163	M	2016-04-26T08:44:12Z	2016-04-29T00:00:00Z	29	NOVA PALESTINA	0	0	0	0	0	1	Yes
566654781423437	5634718	F	2016-04-28T11:33:51Z	2016-04-29T00:00:00Z	22	NOVA PALESTINA	1	0	0	0	0	0	No
911394617215919	5636249	M	2016-04-28T14:52:07Z	2016-04-29T00:00:00Z	28	NOVA PALESTINA	0	0	0	0	0	0	No
99884723334928	5633951	F	2016-04-28T10:06:24Z	2016-04-29T00:00:00Z	54	NOVA PALESTINA	0	0	0	0	0	0	No
99948393975	5620206	F	2016-04-26T08:47:27Z	2016-04-29T00:00:00Z	15	NOVA PALESTINA	0	0	0	0	0	1	No
84574392942817	5633121	M	2016-04-28T08:51:47Z	2016-04-29T00:00:00Z	50	NOVA PALESTINA	0	0	0	0	0	0	No
14794966191172	5633460	F	2016-04-28T09:28:57Z	2016-04-29T00:00:00Z	40	CONQUISTA	1	0	0	0	0	0	Yes
17135378245248	5621836	F	2016-04-26T10:54:18Z	2016-04-29T00:00:00Z	30	NOVA PALESTINA	1	0	0	0	0	1	No
7223289184215	5640433	F	2016-04-29T10:43:14Z	2016-04-29T00:00:00Z	46	DA PENHA	0	0	0	0	0	0	No
622257462899397	5626083	F	2016-04-27T07:51:14Z	2016-04-29T00:00:00Z	30	NOVA PALESTINA	0	0	0	0	0	0	Yes
12154843752835	5628338	F	2016-04-27T10:50:45Z	2016-04-29T00:00:00Z	4	CONQUISTA	0	0	0	0	0	0	Yes
863229818887631	5616091	M	2016-04-25T13:29:16Z	2016-04-29T00:00:00Z	13	CONQUISTA	0	0	0	0	0	1	Yes
213753979425692	5634142	F	2016-04-28T10:27:05Z	2016-04-29T00:00:00Z	46	CONQUISTA	0	0	0	0	0	0	No
8734857996885	5641780	F	2016-04-29T14:19:19Z	2016-04-29T00:00:00Z	65	TABUAZEIRO	0	0	0	0	0	0	No
5819369978796	5624020	M	2016-04-26T15:04:17Z	2016-04-29T00:00:00Z	46	CONQUISTA	0	1	0	0	0	1	No
25787851512	5641781	F	2016-04-29T14:19:42Z	2016-04-29T00:00:00Z	45	BENTO FERREIRA	0	1	0	0	0	0	No
12154843752835	5628345	F	2016-04-27T10:51:45Z	2016-04-29T00:00:00Z	4	CONQUISTA	0	0	0	0	0	0	No
5926171692527	5642400	M	2016-04-29T15:48:02Z	2016-04-29T00:00:00Z	51	SÃO PEDRO	0	0	0	0	0	0	No
1225776163665	5642186	F	2016-04-29T15:16:29Z	2016-04-29T00:00:00Z	32	SANTA MARTHA	0	0	0	0	0	0	No

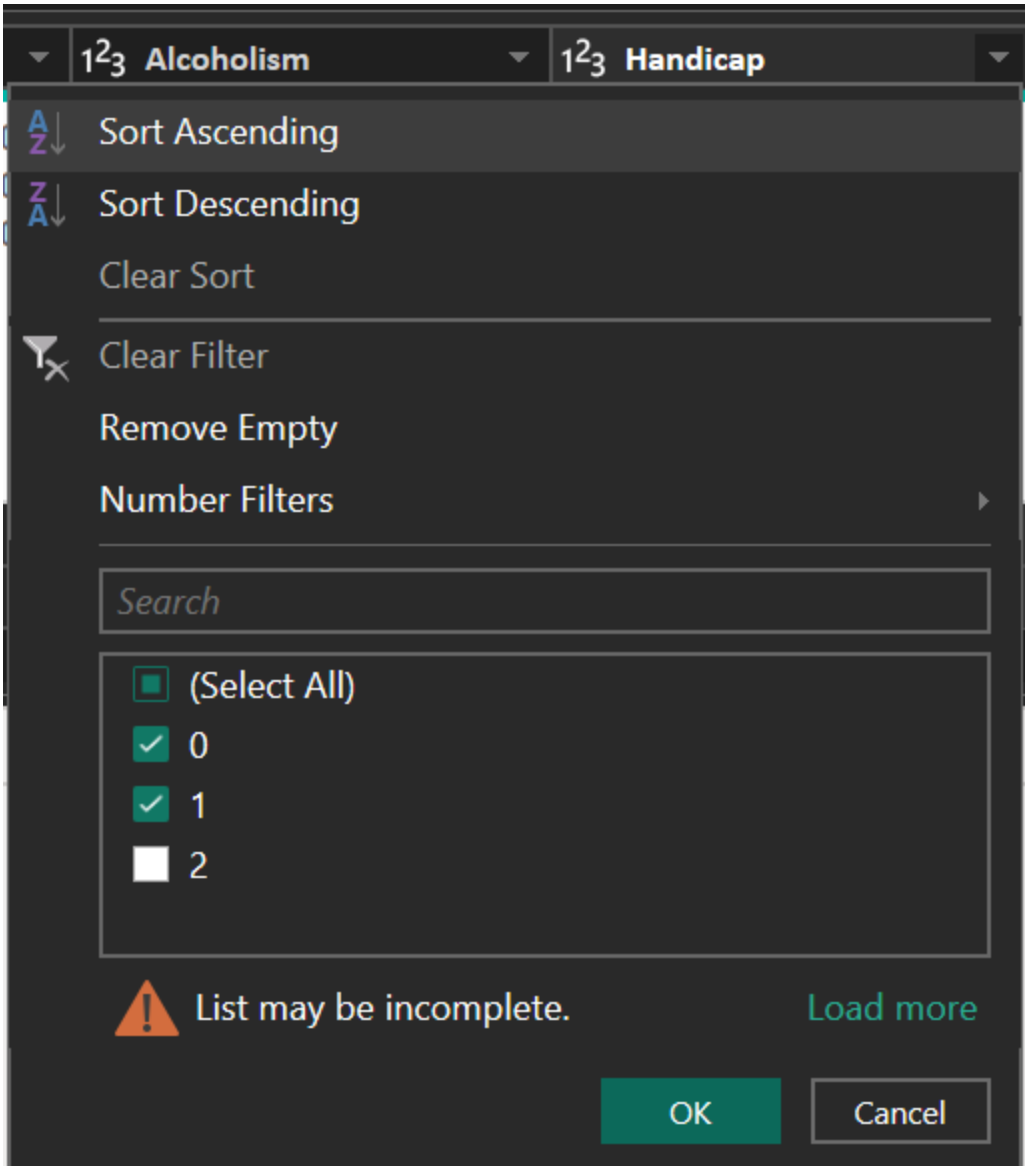
Part B: Data cleaning and transformations

Data Cleaning

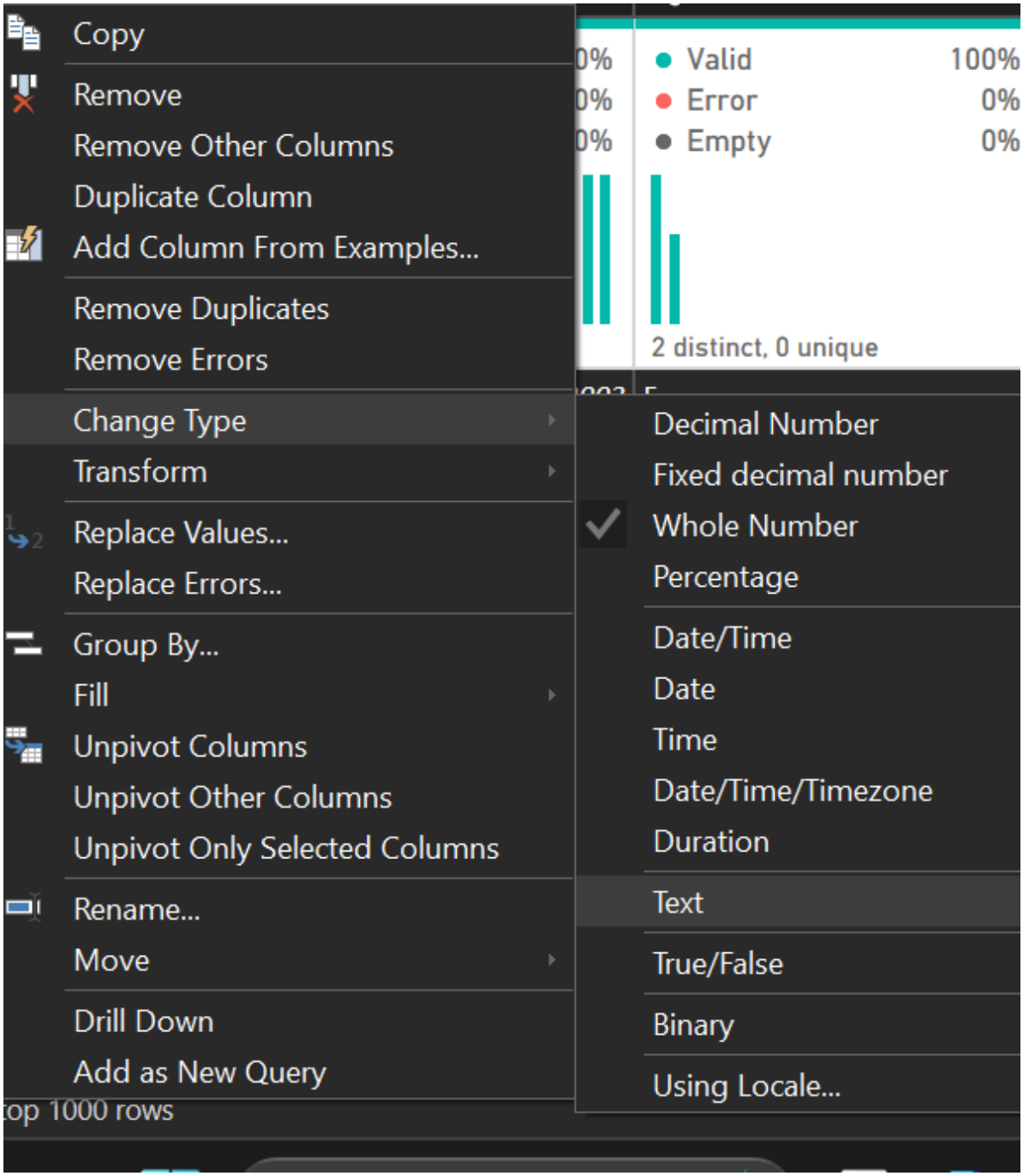
- Before cleaning



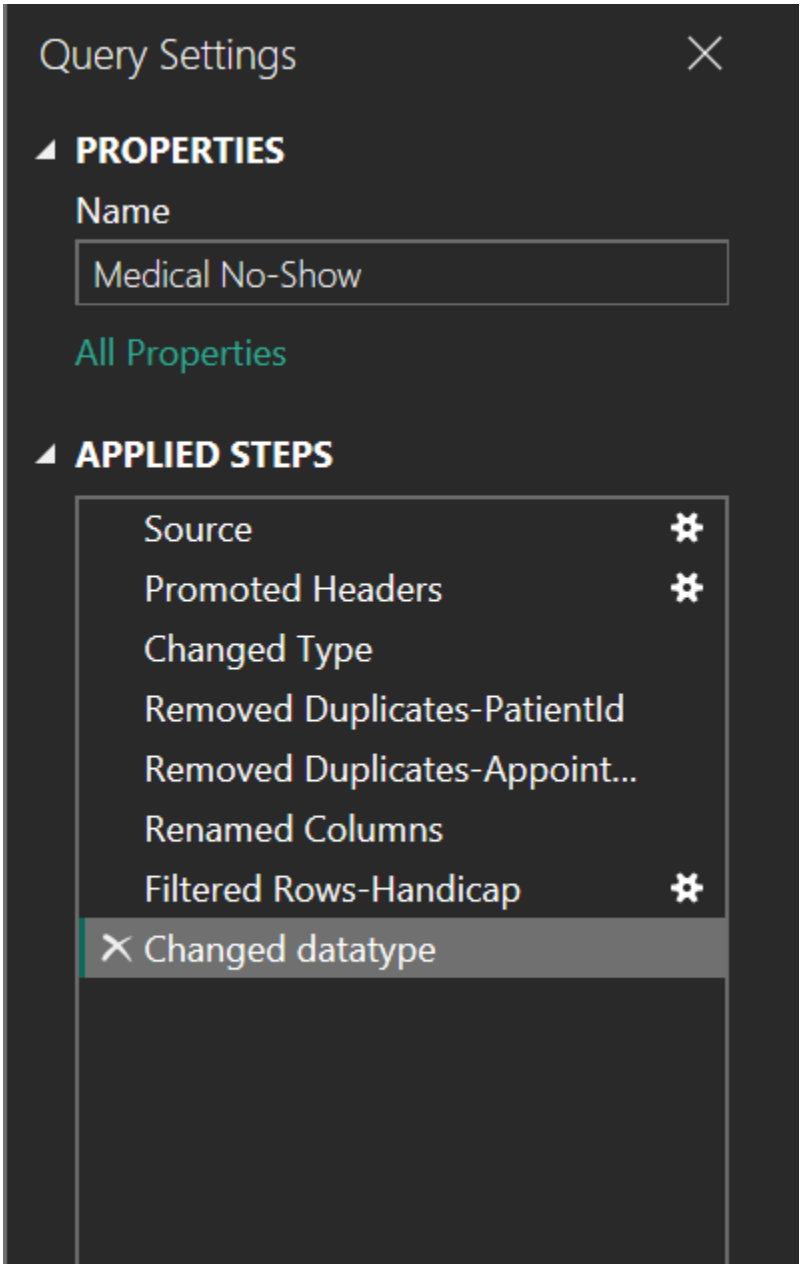
- Removed 2 in Handicapped columns since it is yes(1) or no(0)



- Changed PatientID and AppointmentId datatype to `text` from whole number - you do not perform mathematical operations on IDs


Copy
Remove
Remove Other Columns
Duplicate Column
Add Column From Examples...
Remove Duplicates
Remove Errors
Change Type
Transform
Replace Values...
Replace Errors...
Group By...
Fill
Unpivot Columns
Unpivot Other Columns
Unpivot Only Selected Columns
Rename...
Move
Drill Down
Add as New Query
Valid 100%
Error 0%
Empty 0%
2 distinct, 0 unique
Decimal Number
Fixed decimal number
Whole Number
Percentage
Date/Time
Date
Time
Date/Time/Timezone
Duration
Text
True/False
Binary
Using Locale...
op 1000 rows

- After cleaning

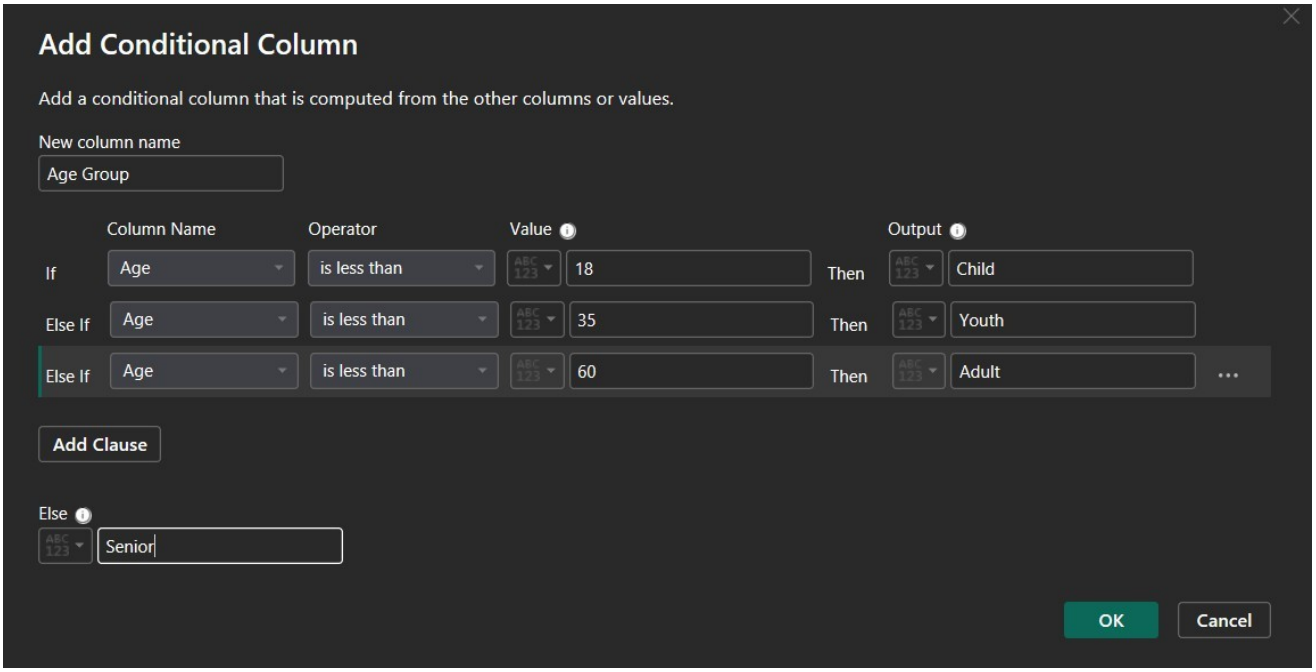


Data Transformation

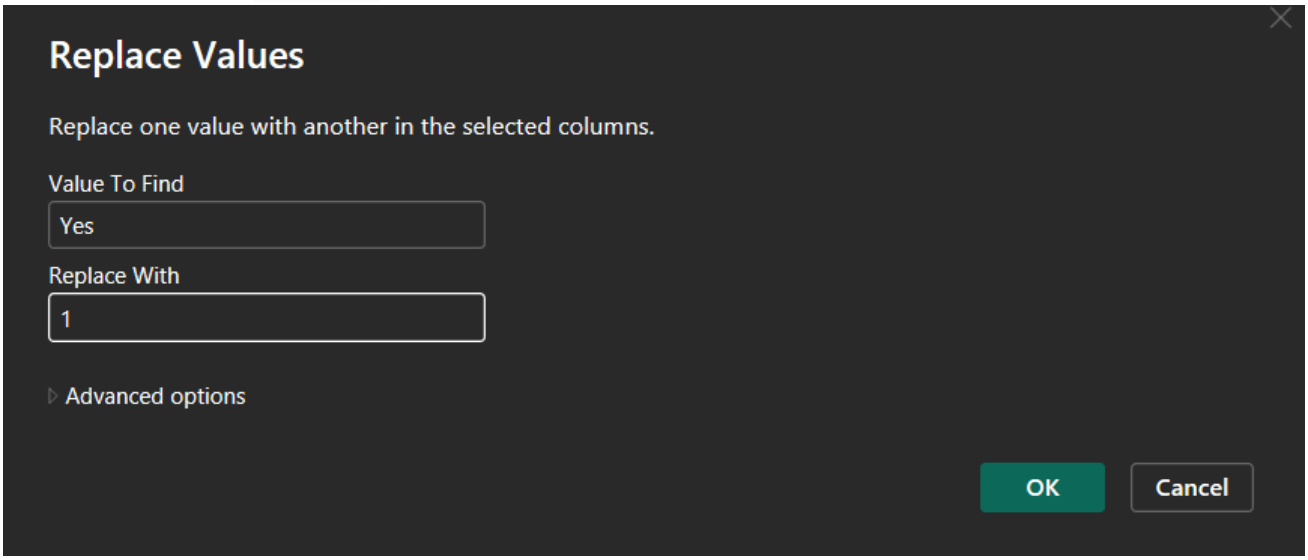
- Extract date part - This creates a new column showing Monday, Tuesday, etc., which is crucial for analyzing no-show trends by weekday.

Scheduled Month Name	Scheduled Day Name	Scheduled Time
Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%
4 distinct, 0 unique	5 distinct, 0 unique	971 distinct, 943 unique
April	Friday	21:38:08
April	Friday	19:08:27
April	Friday	19:19:04
April	Friday	20:29:31
April	Friday	19:07:23
April	Wednesday	11:36:51
April	Wednesday	18:05:12
April	Wednesday	18:39:58
April	Friday	11:02:16
April	Wednesday	15:48:25
April	Wednesday	17:58:11
April	Tuesday	11:44:12
April	Thursday	14:33:51
April	Thursday	17:52:07
April	Thursday	13:06:24
April	Tuesday	11:47:27

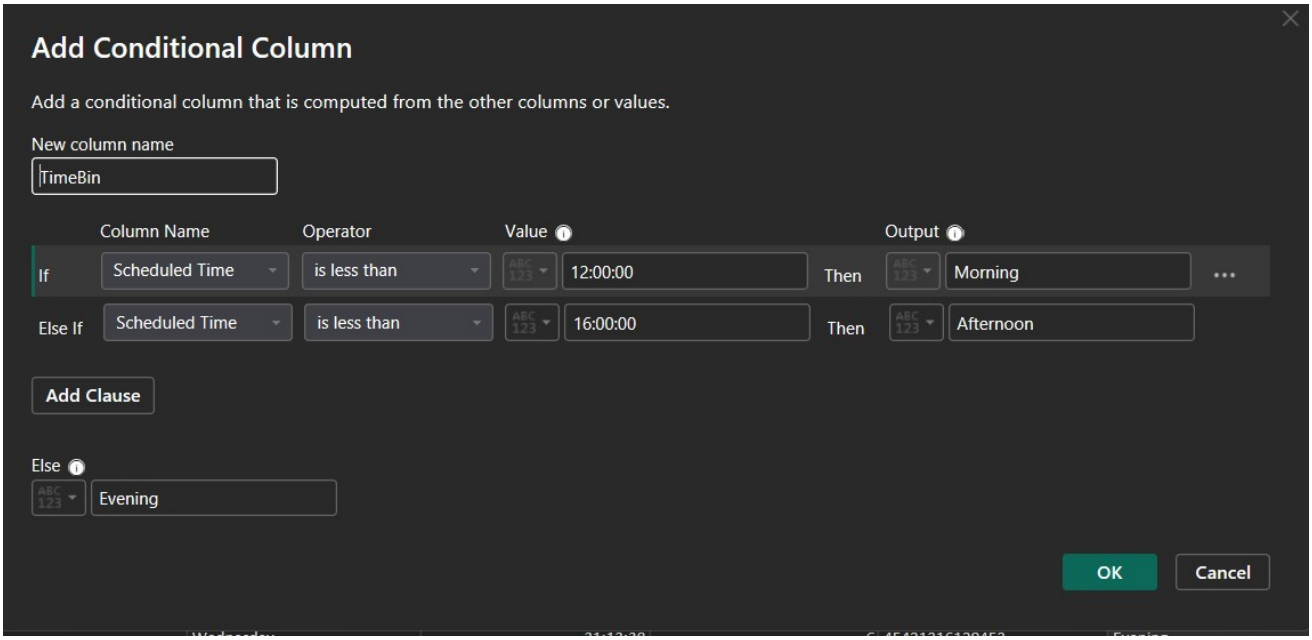
- Age-bins - grouping ages



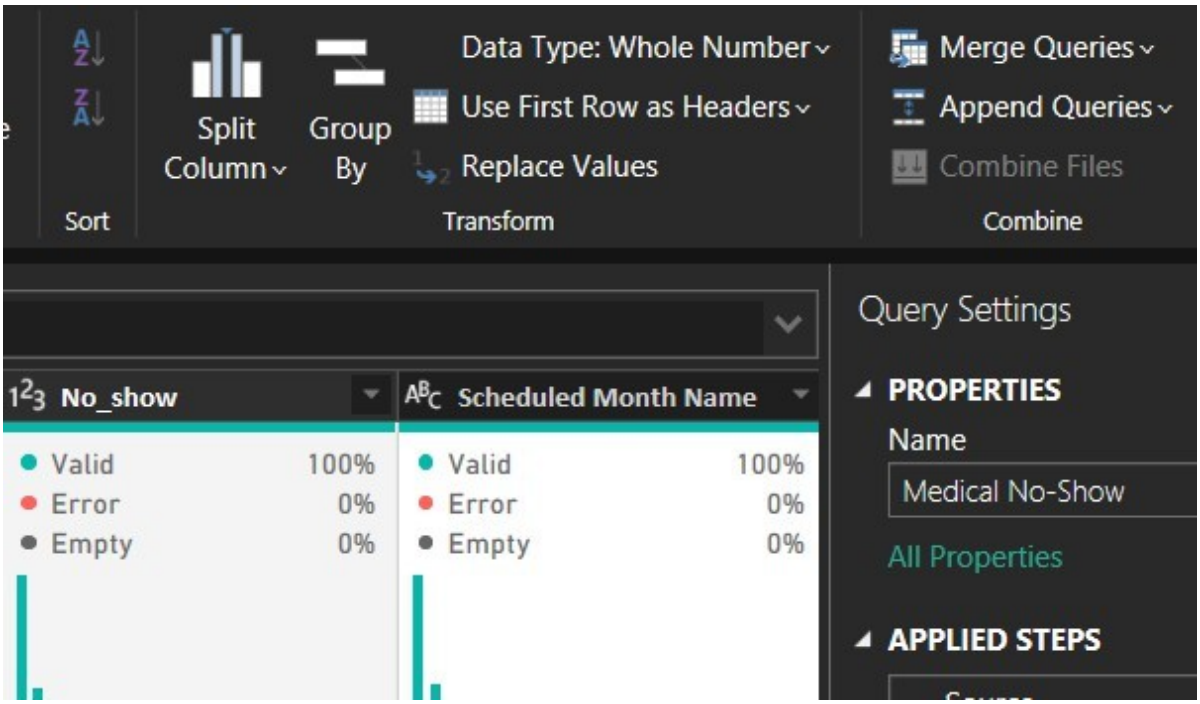
- Replace values - `No-show` with 1 for yes and 0 for no



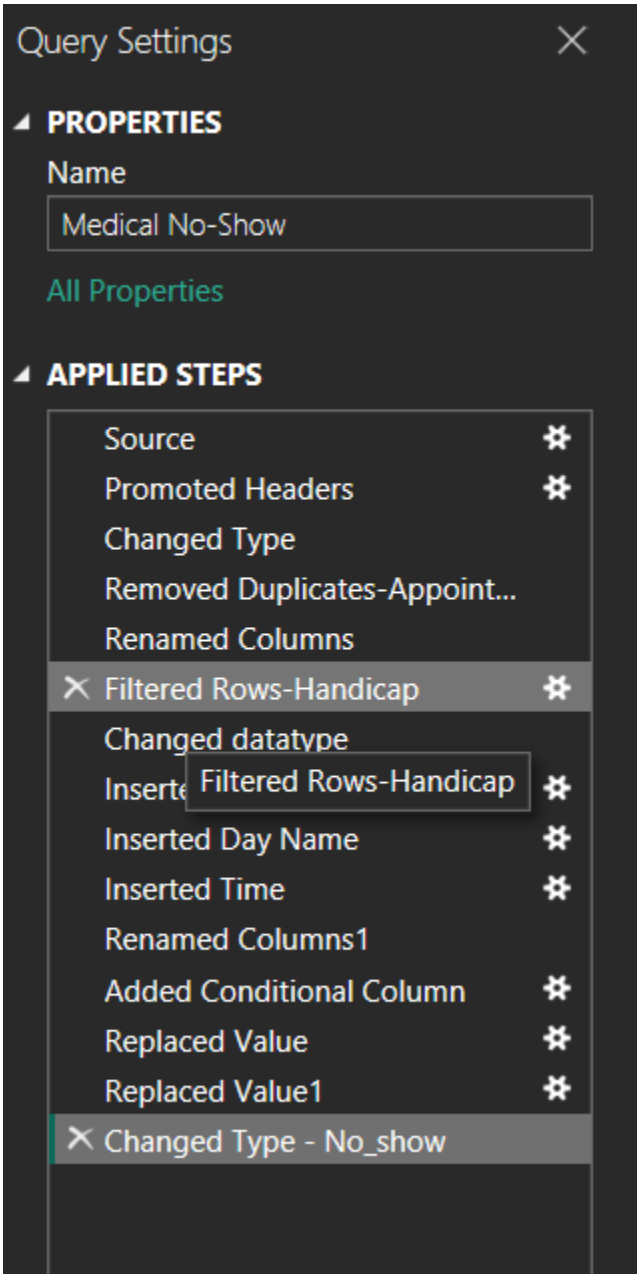
- Time bins



- Changed data type

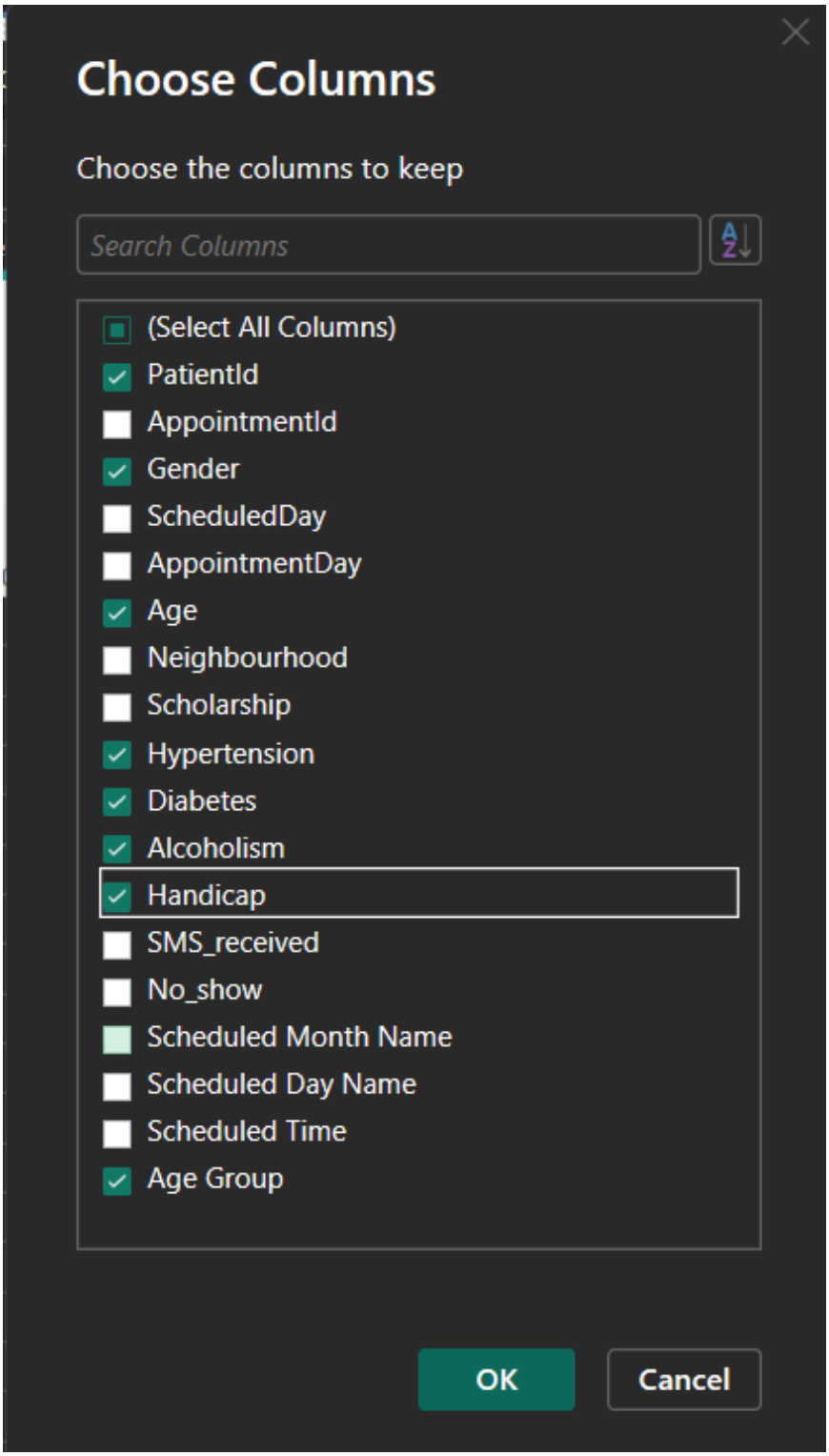


- After transformations

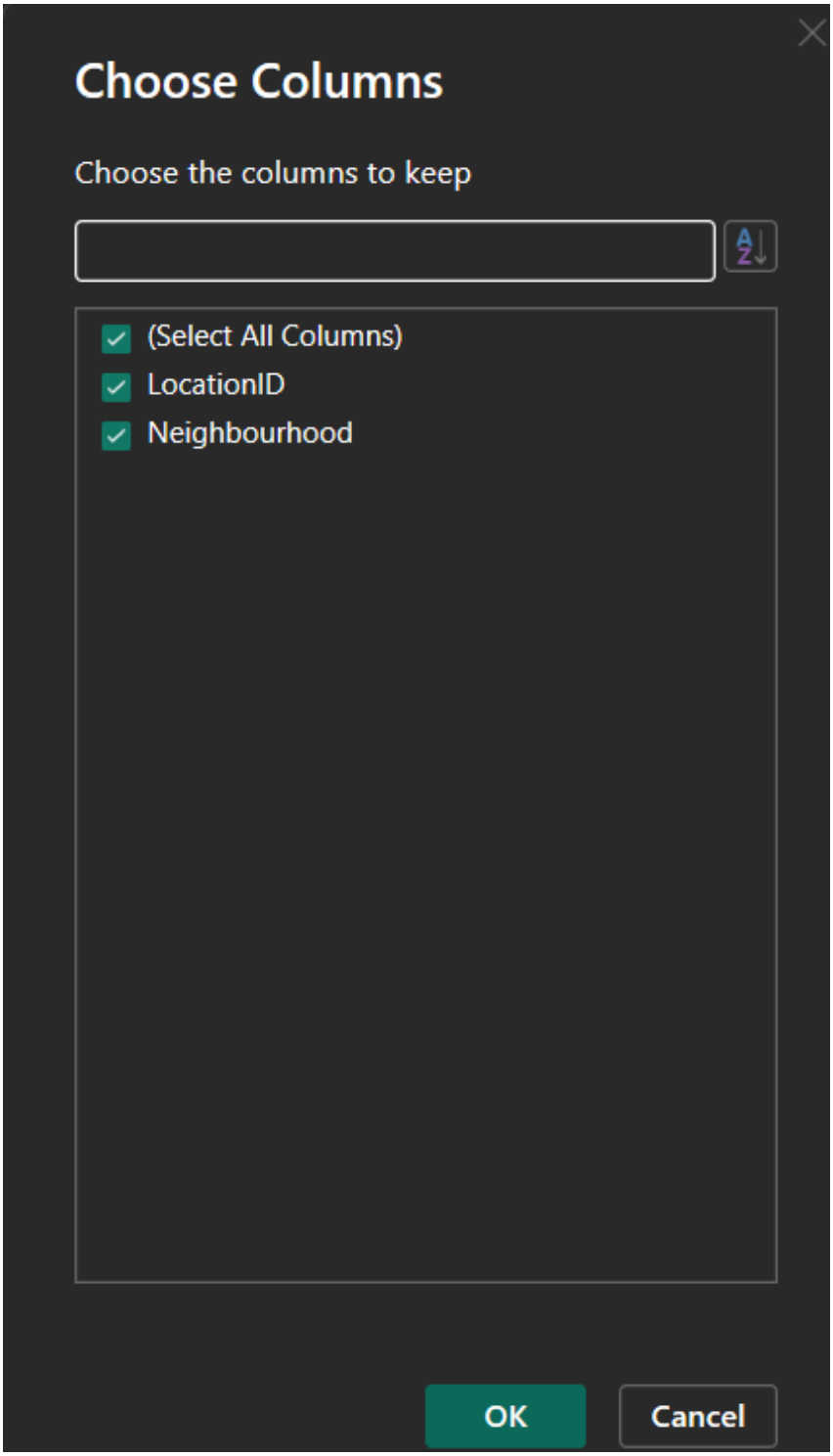


Building a star schema

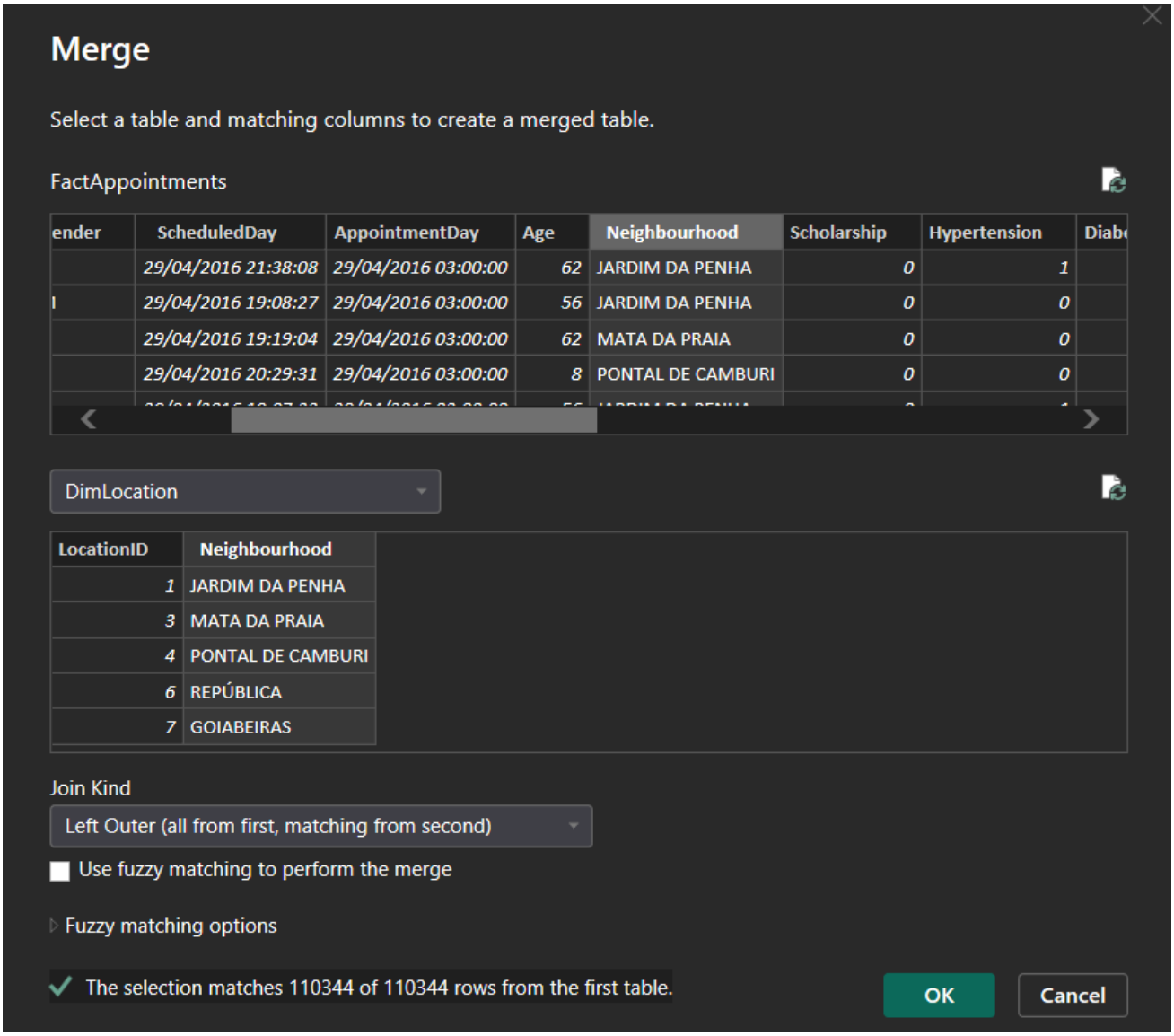
- Create DimPatient

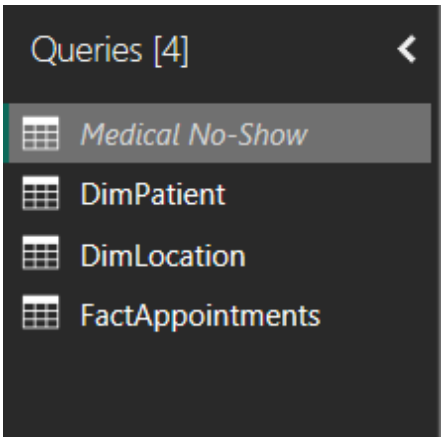


- DimLocation

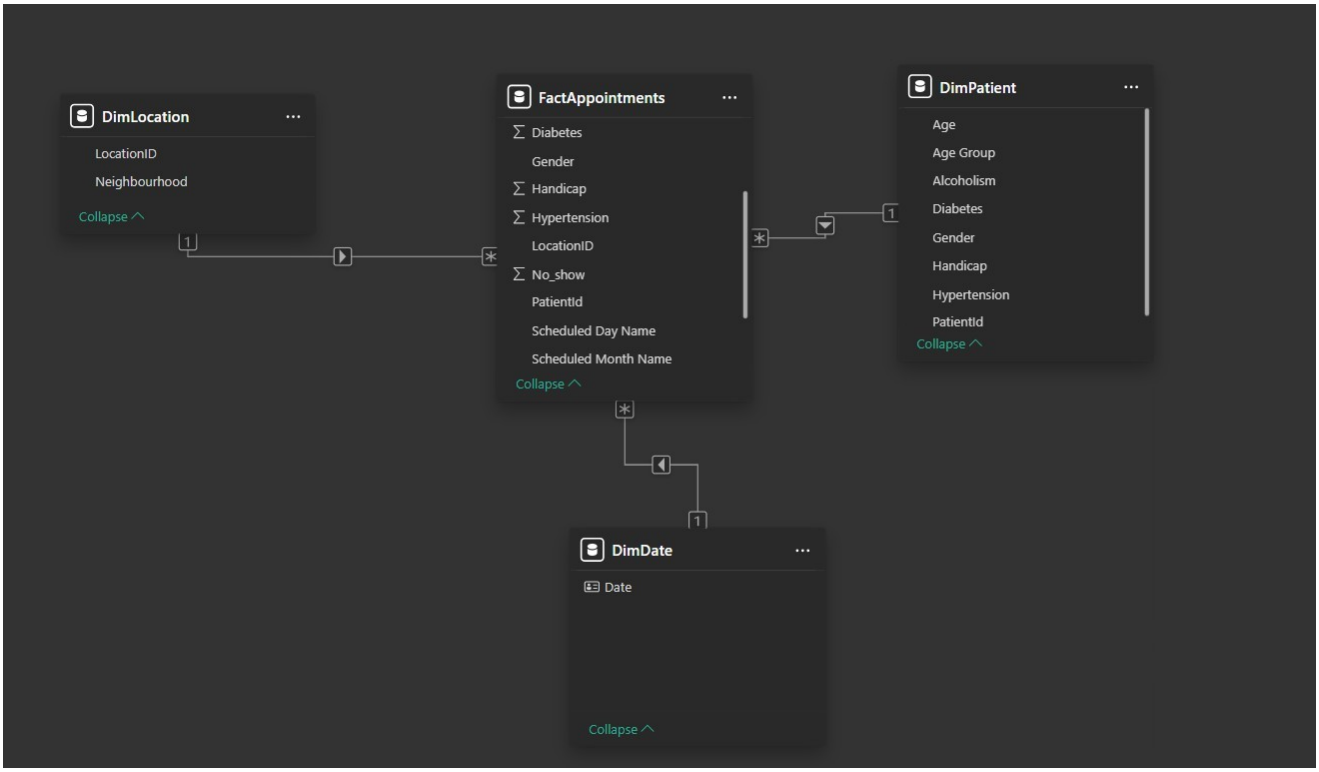


- Merge queries





- Relationship between tables



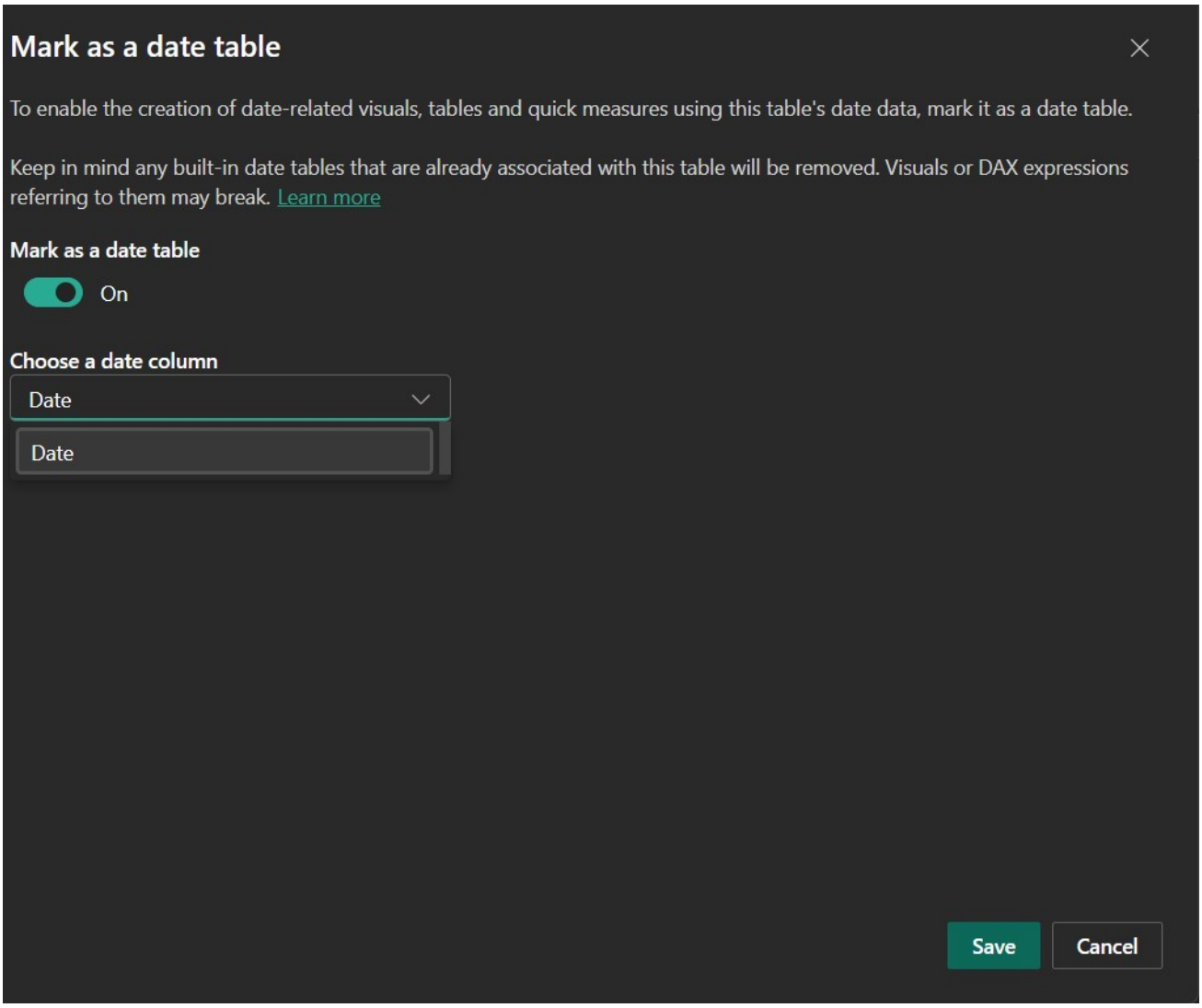
Part C: Data Modeling

- Creating a Date table

```
DimDate = CALENDAR(MIN(FactAppointments[AppointmentDate]),
MAX(FactAppointments[AppointmentDate]))
```

```
DimDate = CALENDAR(MIN(FactAppointments[AppointmentDay]), MAX(FactAppointments[AppointmentDay]))
```

- Marking as data

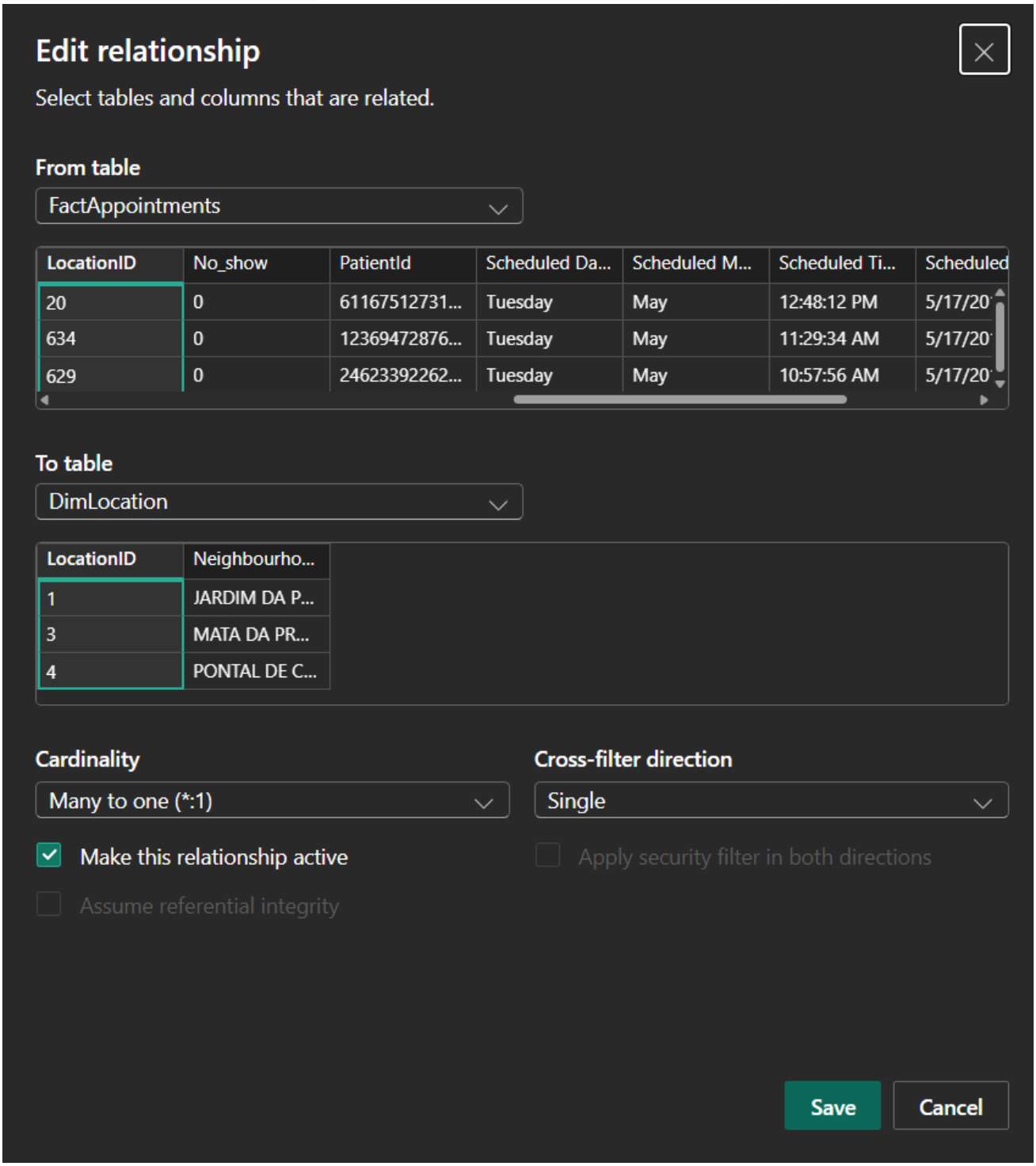


Identify and Justify Tables

- **Fact Table:** `FactAppointments`. Justification: It contains the quantitative events (the appointments) and the measurable target variable (`No_Show`).
- **Dimension Tables:** `DimPatient`, `DimLocation`, `DimDate`. Justification: They contain the descriptive text attributes (demographics, neighborhood names, time periods) used to filter, slice, and group the metrics in the fact table.

*Relationships, Cardinality, and Cross-filter Direction

- `DimPatient[PatientId]` (1) to `FactAppointments[PatientId]` (*)
- `DimLocation[LocationID]` (1) to `FactAppointments[LocationID]` (*)



- `DimDate[Date]` (1) to `FactAppointments[AppointmentDate]` (*). This is your primary, active relationship.

Edit relationship

Select tables and columns that are related.

From table

FactAppointments

Age	Age Group	Alcoholism	Appointment...	AppointmentId	Diabetes	Gender
0	Child	0	5/17/2016 3:0...	5707568	0	M
0	Child	0	5/17/2016 3:0...	5706553	0	M
0	Child	0	5/17/2016 3:0...	5706176	0	M

To table

DimDate

Date
4/29/2016 12:...
4/30/2016 12:...
5/1/2016 12:0...

Cardinality

Many to one (*:1)

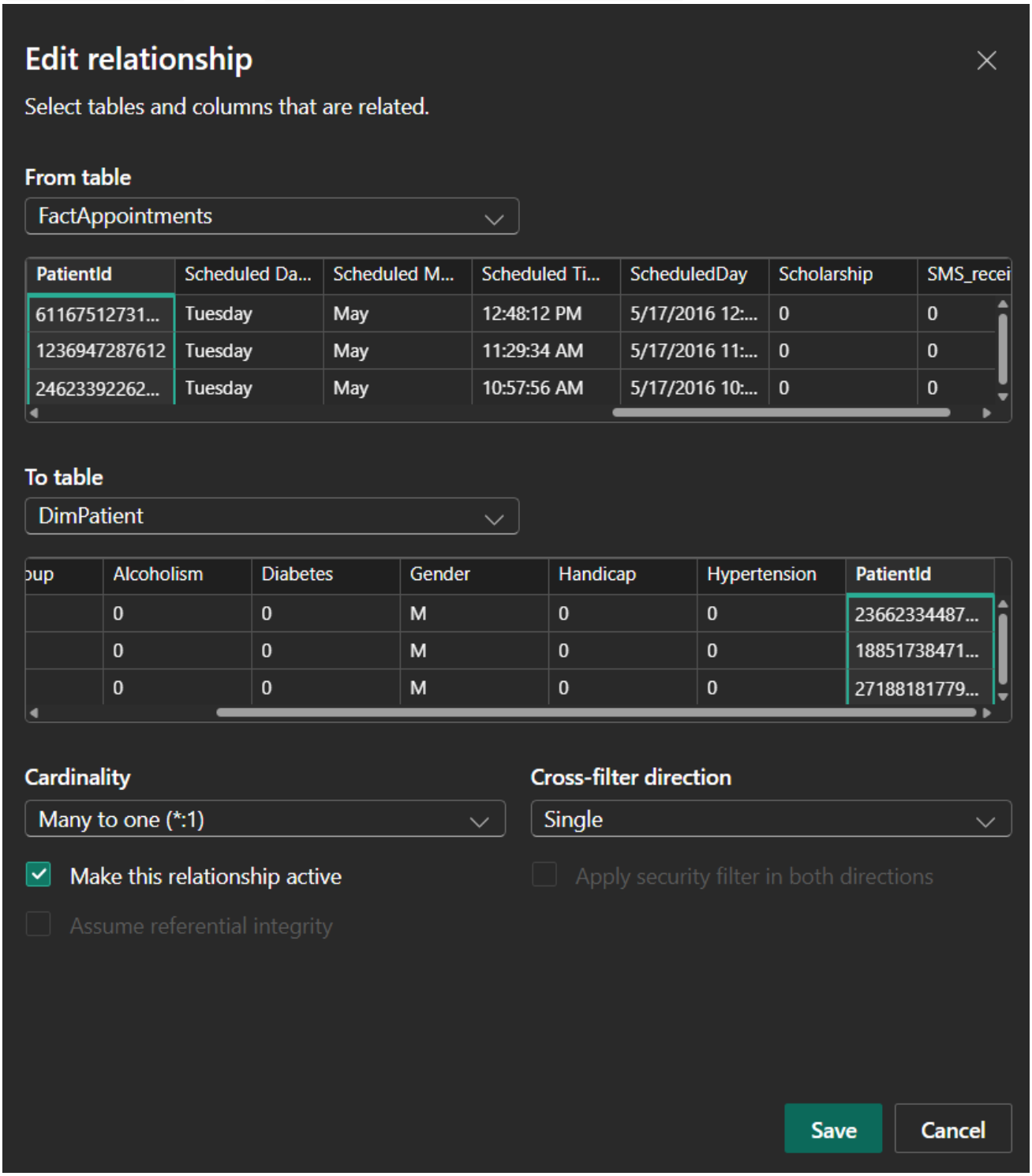
Cross-filter direction

Single

- [x] Make this relationship active
- [ ] Assume referential integrity
- [ ] Apply security filter in both directions

Save Cancel

- `DimDate[Date]` (1) to `FactAppointments[ScheduledDate]` (*).



Model Structure Efficiency

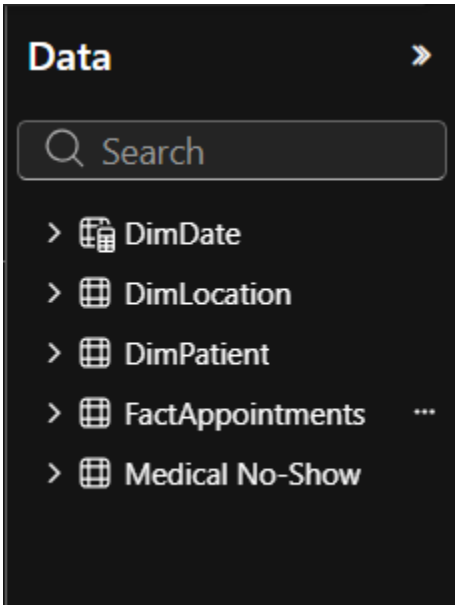
This structure supports efficient reporting because it forms a clean star schema. The single-directional 1:* relationships ensure filters flow downhill from dimensions to facts without creating circular dependencies. It reduces data redundancy, compresses the file size, and allows DAX engine queries to run significantly faster than they would on a flat table.

Explanation of Each Table's Role

- **FactAppointments (Fact Table):** Stores the transactional events (individual appointments) and the measurable target variable (`No_Show`). It contains foreign keys used to link to the surrounding dimension tables.
- **DimPatient (Dimension Table):** Stores descriptive attributes about individual patients, including demographics (Gender, Age Group) and health conditions (Hypertension, Diabetes). It is used to slice and filter fact table metrics by patient profiles.
- **DimLocation (Dimension Table):** Stores geographic data (`Neighbourhood`). It is used to analyze spatial distributions and identify locations with high no-show rates.
- **DimDate (Dimension Table):** A continuous calendar table required for Power BI time intelligence functions. It enables the analysis of trends over days, weeks, and months.

Relationship Description The model utilizes a star schema optimized for Power BI. All relationships flow downstream from dimensions to the central fact table.

- **DimPatient to FactAppointments:** A 1-to-many (1:*) relationship linked via `PatientId` with a single cross-filter direction.
- **DimLocation to FactAppointments:** A 1-to-many (1:*) relationship linked via `LocationID` with a single cross-filter direction.
- **DimDate to FactAppointments (Active):** A 1-to-many (1:*) relationship linked via the `Date` and `AppointmentDate` columns with a single cross-filter direction.
- **DimDate to FactAppointments (Inactive):** A 1-to-many (1:*) relationship linked via the `Date` and `ScheduledDate` columns. This is kept inactive to prevent circular dependencies while allowing activation via DAX (`USERELATIONSHIP`) if analysis based on scheduling date is required.



Part D: DAX Measures and Calculated columns

Calculated columns

1. WaitTimeDays

```
WaitTimeDays = DATEDIFF(FactAppointments[ScheduledDate],
FactAppointments[AppointmentDate], DAY)
```

```
WaitTimeDays = DATEDIFF(FactAppointments[ScheduledDay], FactAppointments[AppointmentDay], DAY)
```

- **What it calculates:** The number of days between when the patient booked the appointment and the actual appointment date.
- **Decision-making value:** Helps identify if patients booking far in advance are more likely to miss their appointments compared to same-day bookings.

2. Risk Level

```
Risk Level =
SWITCH(
    TRUE(),
    FactAppointments[WaitTimeDays] == 0, "Low Risk (Same Day)",
    FactAppointments[WaitTimeDays] <= 7, "Medium Risk (1-7 Days)",
    "High Risk (8+ Days)"
)
```

```
Risk Level =
SWITCH(
    TRUE(),
    FactAppointments[WaitTimeDays] == 0, "Low Risk (Same Day)",
    FactAppointments[WaitTimeDays] <= 7, "Medium Risk (1-7 Days)",
    "High Risk (8+ Days)"
)
```

- ***What it calculates:** Categorizes the appointment into risk buckets based on the wait time using conditional `SWITCH` logic.

- **Decision-making value:** Allows administrators to segment patients and target high-risk (8+ days) appointments with automated SMS reminders.

DAX Measures

1. Total Appointments (Count)

```
Total Appointments = COUNTROWS(FactAppointments)
```

```
Total Appointments = COUNTROWS(FactAppointments)
```

- **What it calculates:** The total number of appointment records in the dataset.
- **Decision-making value:** Provides the baseline volume of clinic activity required to contextualize all other metrics.

2. Total No-Shows (Total Amount)

```
Total No-Shows = SUM(FactAppointments[No_Show])
```

```
Total No-Shows = SUM(FactAppointments[No_Show])
```

- **What it calculates:** Adds up the `1`s in the `No_Show` column.
- **Decision-making value:** Quantifies the exact number of missed appointments, showing the raw scale of the problem.

3. Distinct Patients (Distinct Count)

```
Distinct Patients = DISTINCTCOUNT(FactAppointments[PatientId])
```

```
Distinct Patients = DISTINCTCOUNT(FactAppointments[PatientId])
```

- **What it calculates:** The number of unique individuals who visited the clinic, ignoring multiple visits by the same person.
- **Decision-making value:** Shows whether no-shows are driven by a large group of people missing one appointment, or a small group of repeat offenders missing multiple.

4. No-Show Rate % (Percentage Contribution)

```
No-Show Rate = DIVIDE([Total No-Shows], [Total Appointments], 0)
```

```
No-Show Rate = DIVIDE([Total No-Shows], [Total Appointments], 0)
```

- **What it calculates:** The ratio of missed appointments to total booked appointments.
- **Decision-making value:** This is the primary Key Performance Indicator (KPI). It normalizes performance so you can compare different neighborhoods or months fairly regardless of total volume.

5. Average Wait Time (Average Metric)

```
Average Wait Time = AVERAGE(FactAppointments[WaitTimeDays])
```

```
Average Wait Time = AVERAGE(FactAppointments[WaitTimeDays])
```

- **What it calculates:** The mean number of days patients wait for their appointments.
- **Decision-making value:** Highlights operational bottlenecks. If this number climbs, the clinic is struggling to meet patient demand promptly.

6. YTD Total Appointments (Time Intelligence)

```
YTD Total Appointments = TOTALYTD([Total Appointments], DimDate[Date])
```

```
YTD Total Appointments = TOTALYTD([Total Appointments], DimDate[Date])
```

- **What it calculates:** The cumulative sum of appointments from January 1st to the current context date.
- **Decision-making value:** Allows management to track cumulative annual volume against yearly operational targets.

7. MoM No-Show Growth % (Change Over Time)

```
MoM No-Show Growth =
VAR PrevMonth = CALCULATE([Total No-Shows], DATEADD(DimDate[Date], -1,
MONTH))
RETURN DIVIDE([Total No-Shows] - PrevMonth, PrevMonth, 0)
```

```
MoM No-Show Growth =
VAR PrevMonth = CALCULATE([Total No-Shows], DATEADD(DimDate[Date], -1, MONTH))
RETURN DIVIDE([Total No-Shows] - PrevMonth, PrevMonth, 0)
```

- **What it calculates:** The percentage increase or decrease in raw no-shows compared to the previous month.
- **Decision-making value:** Acts as an early warning system. A positive growth percentage indicates the no-show problem is worsening and requires immediate intervention.

8. Location No-Show Rank (Ranking)

```
Location Rank = RANKX(ALL(DimLocation[Neighbourhood]), [Total No-Shows], ,
DESC, Dense)
```

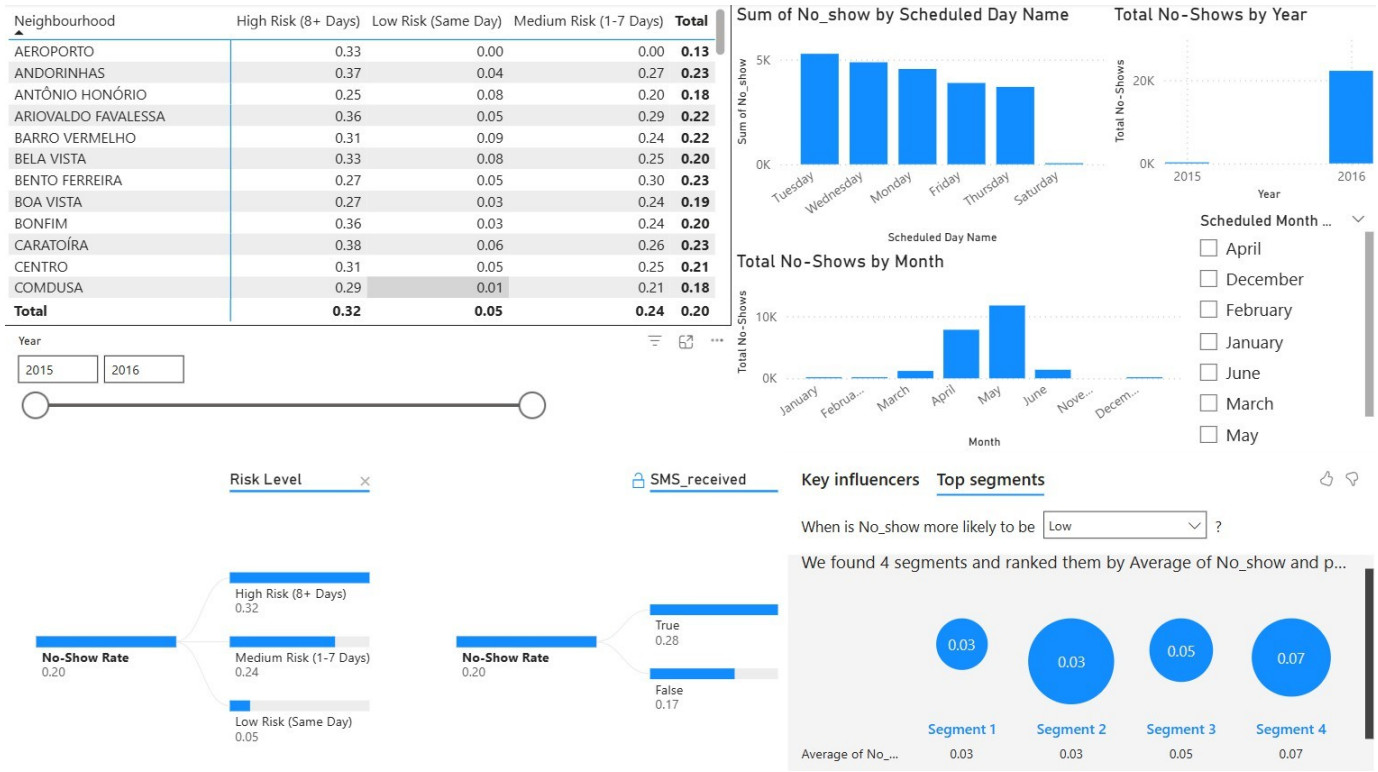
```
Location Rank = RANKX(ALL(DimLocation[Neighbourhood]), [Total No-Shows], , DESC, Dense)
```

- **What it calculates:** Assigns a rank (1, 2, 3...) to each neighborhood based on its total number of no-shows.
- **Decision-making value:** Directs geographic resource allocation. It tells clinic staff exactly which neighborhood to target first for community outreach or targeted reminders.

Part E: Dashboard Design and Visualization (20 Marks)

Page1: Executive summary

Page2: Detailed Analysis



Page3: Insights and performance monitor

- **Explanation of Design Choices**: The dashboard utilizes a top-down information architecture. High-level KPIs are positioned at the top left, matching natural reading patterns. A minimal, high-contrast color theme is applied to avoid visual clutter, using a designated warning color (red) exclusively for negative metrics like no-show volume. Visuals are aligned on a strict grid layout to maintain professional aesthetics. Slicers are positioned uniformly across pages to establish a predictable user interface.

- **Explanation of User Interactions**: The dashboard relies on bidirectional cross-filtering. Selecting a specific demographic category (e.g., clicking the 'Senior' bar) dynamically updates all other visuals on the page to reflect that cohort's metrics. Slicers for Date and Location are synchronized across all three pages to maintain context during navigation. The Detailed Analysis page utilizes hierarchical drill-down functionality, allowing users to unpack annual data into monthly and daily trends, while the Decomposition Tree enables ad-hoc root cause analysis by letting users expand specific branches of the data model. Tooltips are enabled on all visual elements to provide exact numeric values upon hover.

Part F: Analytical Insights and Business Recommendations

Analytical Insights

1. **Lead Time Correlation:** The decomposition tree and matrix visuals prove a severe correlation between booking lead time and absenteeism. High-risk appointments (booked 8+ days in advance) suffer a 32% no-show rate, while same-day bookings hold a minimal 5% no-show rate.
2. **The SMS Paradox:** The decomposition tree reveals a critical systemic failure: patients who received an SMS reminder exhibit a higher no-show rate (28%) than those who did not (17%). The current SMS strategy is either targeting the wrong appointments or is timed ineffectively.
3. **Demographic Concentration:** Adult females drive the majority of absenteeism. Females account for 65.38% of all missed appointments, and the "Adult" category dwarfs Youth, Children, and Seniors in total no-show volume.
4. **Time-of-Day Attrition:** Evening appointments experience the highest absolute volume of no-shows compared to mornings and afternoons, pointing to end-of-day scheduling conflicts or fatigue.
5. **Clinical Profile:** Absenteeism is not driven by chronic health complications. Over 83% of no-shows have no hypertension, 93% have no diabetes, and 98% have no handicap. Healthy individuals are missing appointments due to logistical barriers, not medical ones.

Actionable Recommendations

1. **Implement Predictive Overbooking:** Apply a strict overbooking multiplier to evening slots and appointments booked 8+ days in advance to counteract the predictable 32% revenue/time loss in these specific segments.
2. **Overhaul the SMS Architecture:** Immediately suspend and rebuild the SMS reminder protocol. Shift from automated blanket messaging to targeted reminders sent exactly 24 hours prior, focusing specifically on the high-risk adult female demographic.

3. **Shift Capacity to Same-Day Triage:** Since same-day appointments guarantee a 95% attendance rate, restructure clinic schedules to reserve a larger block of daily capacity for walk-ins and short-notice bookings, reducing the reliance on high-attrition, long-lead scheduling.

Conclusion

This project successfully delivered a comprehensive Business Intelligence solution to analyze and mitigate medical appointment no-shows. By transforming over 110,000 raw medical records into a functional star-schema relational model, we created a robust foundation for advanced analytics. The application of DAX-driven KPIs, such as the No-Show Rate and Risk Level logic, allows for real-time monitoring of clinical efficiency.

The resulting multi-page dashboard provides stakeholders with the tools necessary to identify high-risk scheduling windows and demographic attrition patterns. Implementing the proposed data-driven recommendations—specifically targeted overbooking and the restructuring of the SMS notification architecture—will enable the clinic to optimize resource allocation and improve patient health outcomes.

References

Kaggle: https://www.kaggle.com/datasets/joniarroba/noshowappointments
GitHub:https://github.com/Calvin-Gacheru/Business-Intelligence-And-Visualization/tree/main/End%20Semester%20Project